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HomeWork 2

Solution

1) i) Given

Sample Size $n = 50$

No. of predictors $p = 3$

To build

Multiple Linear Regression } $y = X\beta + \epsilon$
Model

where, Design Matrix X

Response y

Predictors: x_1 (intercept), x_2, x_3

To find

i) OLS estimator,

$$\hat{\beta} = (X^T X)^{-1} X^T y$$



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In the given dataset,

$$y = \begin{bmatrix} -0.344 \\ -0.546 \\ \vdots \\ -0.982 \end{bmatrix}, \quad X = \begin{bmatrix} 1 & -0.626 & 0.398 \\ 1 & 0.184 & -0.612 \\ \vdots & \vdots & \vdots \\ 1 & 0.881 & -0.473 \end{bmatrix}$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

Substituting the values, in the formula,
we get,

$$\hat{\beta} = \left(\begin{bmatrix} 1 & 1 & \dots & 1 \\ -0.626 & 0.184 & \dots & 0.881 \\ 0.398 & -0.612 & \dots & -0.473 \end{bmatrix} \begin{bmatrix} 1 & -0.626 & 0.398 \\ 1 & 0.184 & -0.612 \\ \vdots & \vdots & \vdots \\ 1 & 0.881 & -0.473 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 & \dots & 1 \\ -0.626 & 0.184 & \dots & 0.881 \\ 0.398 & -0.612 & \dots & -0.473 \end{bmatrix} \begin{bmatrix} -0.344 \\ -0.546 \\ \vdots \\ -0.982 \end{bmatrix}$$



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On Solving this, we get,

$$\hat{\beta} = \begin{pmatrix} -0.5097 \\ -0.5340 \\ 0.0873 \end{pmatrix}$$

ii) To find, Distribution of $\hat{\beta}_3$

Under Standard Regression Assumptions,

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})$$

$$\Rightarrow \hat{\beta}_3 \sim N(\beta_3, \sigma^2 [(X^T X)^{-1}]_{33})$$

On solving this programmatically,
we get,

$$\hat{\beta}_3 \sim N(\beta_3, 0.001261)$$

~~$$\hat{\beta}_3 \sim N(\beta_3, 0.001261)$$~~

~~$$\hat{\beta}_3 \sim N(0.0873, 0.001261)$$~~



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1) 2) Given $\beta_3 = 0$

$$\alpha = 0.01$$

To perform F-test for H_0

Hypothesis: $H_0: \beta_3 = 0$; $H_a: \beta_3 \neq 0$

As this is single linear restriction, F-test uses

$$F = \frac{(SSR_{\text{reduced}} - SSR_{\text{full}}) / 1}{SSE_{\text{full}} / (n - p)}$$

Under H_0 ,

$$F \sim F_{1, n-p} \sim F_{1, 47}$$

Reject H_0 if,

$$F_{\text{stat}} > F_{1, 47} (0.01)$$

(stat) (critical)

Resolving this programmatically, we get

$$F_{\text{stat}} = 6.04, F_{\text{crit}} = F_{1, 47} (0.01) = 7.21$$

$\Rightarrow F_{\text{stat}} < F_{\text{crit}} \Rightarrow$ Cannot reject H_0

So, there is insufficient evidence to conclude that β_3 is statistically different from zero.



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i) 3) Given $\beta_3 = 0$, $\alpha = 0.01$; Degrees of freedom = 47
To perform, t-test

We know that,

$$\text{Test statistic } t = \frac{\hat{\beta}_3}{\text{SE}(\hat{\beta}_3)} = \frac{0.0873}{0.036} = 2.46 //$$

We know these values from previous derivations.

Calculating p value programmatically,
we get p-value = 0.018 //

Based on t-test, $p > 0.01 \Rightarrow$ Fail to reject H_0

So, Similar to F-test, even using t-test,
 β_3 is not significant at the 1% level. //

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1) 4) Leverage values are the diagonal elements of the hat matrix:

$$H = X(X^T X)^{-1} X^T$$

Substituting the values and solving programmatically we get

Leverage score for

$$\text{Observation 1: } h_{11} = 0.0369$$

$$\text{Observation 2: } h_{22} = 0.0317$$

As $h_{11} > h_{22} \Rightarrow$ Observation 1 is more influential.

Issues with very high leverage:

→ These points can dominate the regression fit

→ Makes estimates unstable

→ May lead to misleading inference if they are outliers in the response

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Q5) Given,

New data point $x_3 = (1, 0.12, 0.56)$

To find,

Predicted value $y = x^T \beta$

Solving this programmatically, we get.

$$y = -0.525 //$$

95% Confidence Interval:

$$(-0.60, -0.45) //$$

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2) 1) Given

p -dimensional multiple linear model

$$y = X\beta + \epsilon$$

$$X^T X = D$$

$$D = \text{diag}(d_j)$$

$$d_1 = \eta \Rightarrow \frac{1}{\eta} \text{ (very large)}$$

(small)

We know that

$$\text{OLS estimator } \hat{\beta} = (X^T X)^{-1} X^T y$$

Since $d_1 = \eta \approx 0$

$$\Rightarrow (X^T X)^{-1}_{11} = \frac{1}{\eta} \approx \infty$$

$\Rightarrow$ Variance of $\hat{\beta}_1$ becomes very large

$\Rightarrow$ Estimates become numerically unstable

$\Rightarrow$ This leads to multicollinearity issue //

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2)2) Given

Ridge estimator,

$$\hat{\beta}_R = (X^T X + \lambda I)^{-1} X^T y$$

The variance of Ridge estimator for β_1 ,

$$\text{Var}(\hat{\beta}_{R1}) = \sigma^2 \frac{\eta}{(\eta + \lambda)^2}$$

$\Rightarrow$ Even when $\eta \rightarrow 0$, variance remains finite

$\Rightarrow$ So, ridge estimator stabilizes estimation

~~2)3) We know that~~

~~$$E(\hat{\beta}_R) = (X^T X + \lambda I)^{-1} X^T X \beta$$~~

~~Bias of Ridge estimator, (upon derivation)~~

~~$$\text{Bias} = E(\hat{\beta}_R) - \beta = -\lambda (X^T X + \lambda I)^{-1} \beta$$~~



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~~2) 4)~~ 2) 3) We know that,

$$E[\hat{\beta}_R] = (X^T X + \lambda I)^{-1} X^T X \beta$$

Bias of Ridge Estimator,

$$\text{Bias}(\hat{\beta}_R) = E[\hat{\beta}_R] - \beta$$

$$= (X^T X + \lambda I)^{-1} X^T X \beta - \beta$$

$$= [(X^T X + \lambda I)^{-1} X^T X - I] \beta$$

We also know that,

$$X^T X = (X^T X + \lambda I) - \lambda I$$

Substituting in this equation, we get.

$$\text{Bias}(\hat{\beta}_R) = [I - \lambda (X^T X + \lambda I)^{-1} - I] \beta$$

$$\Rightarrow \text{Bias}(\hat{\beta}_R) = -\lambda (X^T X + \lambda I)^{-1} \beta$$

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2) 4) When $\lambda \rightarrow 0$

$\Rightarrow$ Ridge $\rightarrow$ OLS

So, as we know from previous derivations,
Instability remains; large variance

But when $\lambda \rightarrow \infty$,

$\Rightarrow$ Coefficients $\rightarrow 0$

which leads to large bias.

So, optimal λ is required to balance
bias and variance.

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2) 5) We know that,

$$\text{For } \beta_1 : \text{MSE}_{\text{OLS}} = \frac{\sigma^2}{\eta}$$

$$\text{MSE}_{\text{Ridge}} = \frac{\sigma^2 \eta}{(\eta + \lambda)^2} + \left(\frac{\lambda}{\eta + \lambda} \beta_1 \right)^2$$

So, Ridge has smaller MSE when,

$$\lambda < \frac{\sigma^2}{\beta_1^2} \quad \& \quad \lambda \gg \eta$$

Compared to λ ,

$\Rightarrow \eta$ is very small

$\Rightarrow$ True coefficient β_1 is not too large

$\Rightarrow \sigma$ is not too small

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